

**On Faultization:
Generative Pre-Trained Transformer.
What Perturbation Reveals About Pattern Access
in a Minimal Transformer**

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Abstract

We apply faultization, a systematic regime of training-time perturbation, to a minimal transformer (4-layer, 16-dimensional, 4-head character-level GPT), and read the results through Levin’s (2026) proposal that physical systems are interfaces onto a latent space of patterns. Twelve experiments perturb the weights, gradients, and communication channels during training (Exp 1-6) and across multi-phase interventions (Exp 7-12), and measure the effect on a single scalar: final training loss. We adopt a three-scale protocol: $n = 3$ pilot data provides coarse signal, $n = 30$ paired analysis (matched seeds) resolves ambiguity, and $n = 300$ sharpens or dissolves it. The most robust findings are equalities of endpoint loss. Radically different training interfaces reach indistinguishable final loss: cell-view (local layerwise loss, no inter-layer gradient flow) is not significantly different from end-to-end backpropagation ($p = 0.24$ at $n = 30$; $+0.1\%$, $p = 0.53$ at $n = 300$), chimeras assembled from independently trained networks match control final loss (all $p > 0.07$), and a layer reset to random weights retrains to control-level final loss (all $p > 0.17$ at $n = 30$). Final loss is a scalar, so reaching the same value does not establish the same solution, representation, or loss-basin, none of which we measured; these are equalities of endpoint loss, not demonstrated equivalences, and no formal equivalence test was run. The sharpest result is an asymmetry between inactive and corrupted components: after normal training, freezing layers 2-3 costs almost nothing (-0.1% , $p = 0.41$ at $n = 300$), whereas negating their gradients degrades final loss by $+26.3\%$ ($p < 0.0001$, $d = +0.531$ at $n = 300$). Silence is tolerated; sabotage is catastrophic. (A frozen layer stays active in the forward pass; silence here means no further weight updates, not absence.) Gradient degradation is absorbed up to a threshold between $\sigma = 0.01$ ($+0.0\%$, $p = 0.86$) and $\sigma = 0.1$ ($+2.3\%$, $p < 0.0001$); replacing the gradient with its sign alone, which discards magnitude, degrades final loss more ($+4.9\%$, $p < 0.0001$) than moderate noise, so magnitude information matters under this fixed Adam recipe, though this is not a controlled comparison with signSGD. We also report a null we initially mistook for a discovery: gradual noise exposure appears to build tolerance that sudden exposure does not, but the gradual schedule injects only about one-third the cumulative gradient-noise variance of the always-on condition, and the closest dose-matched comparison (gradual vs sudden-half, doses 0.67 vs 1.0) shows no inoculation effect at either scale ($p = 0.35$ at $n = 30$, $p = 0.20$ at $n = 300$). Degradation tracks total noise dose, not the order of exposure, so we do not claim a stress-inoculation free lunch.

1. Introduction

Levin (2026) argues that non-physical patterns are real, discovered, and causal, and that physical systems (embryos, computers, algorithms) are interfaces through which patterns from a latent space manifest, accessing them rather than creating them. On this view evolution exploits them as affordances it does not need to pay for: once you find a voltage-gated ion channel, you have a transistor, and truth tables are yours for free. This paper takes that proposal not as an established ontology but as a working vocabulary, and asks what a minimal transformer receives that was not written into its construction or training: what free lunches it appears to get, meaning capabilities not specified by the optimization objective.

This paper applies that question to a minimal transformer. The methodology is **faultization**: systematic perturbation of a system to reveal what patterns it accesses and how robustly it accesses them. Faultization degrades the interface (the weights, gradients, and communication channels through which the optimizer operates) and observes what happens to the patterns the system was manifesting. Four outcomes are possible:

1. **Pattern manifestation**: the system still reaches its training endpoint (the same final loss) despite the perturbation.
2. **Pattern fidelity**: how much interface degradation the pattern tolerates before access is lost. There exists a threshold below which the pattern manifests cleanly and above which it does not.
3. **Pattern corruption**: the interface actively misleads, inverting the pattern. Rather than failing to access the pattern, the system accesses a corrupted version.
4. **Free lunch**: what the system receives without paying for. The optimizer specifies loss minimization; the system exhibits capabilities (recovery, regeneration, trajectory improvement) that were not specified by the objective. This category demands care: a candidate free lunch must survive controls, and Section 4.9 reports one (apparent stress inoculation) that did not.

We use “free lunch” informally throughout, for a capability the system exhibits that was not written into its training objective. It is not the No-Free-Lunch theorem of Wolpert and Macready (1997), which concerns performance averaged over a class of objective functions and makes no claim about individual capabilities; we borrow only the phrase, not the theorem. Read carefully, a “free lunch” in our sense is never literally free: it is paid for by architecture, overparameterization, auxiliary supervision, and additional training steps, and “free” means only that the capability is absent from the loss the optimizer was handed. This paper looks for such free lunches in a minimal transformer under twelve perturbation protocols.

Transformers are typically studied through their outputs: probing learned representations (Belingov & Glass, 2019), ablating components to measure importance (Michel et al., 2019), or tracing computational circuits (Elhage et al., 2021). These methods characterize what the system has learned or which parts matter. They do not ask what happens when the system is forced to learn under constraint, when its interface is *degraded*. Zhang, Goldstein, and Levin (2024) introduced a different methodology in the context of simple algorithms: rather than analyzing sorting algorithms through their final outputs, they perturbed the algorithms during execution, revealing competencies (fault tolerance, delayed gratification, emergent aggregation) that were invisible during normal operation. The central insight: **perturbation reveals what normal operation conceals**.

The question of whether any finding is genuine requires statistical power. We adopt an explicit three-scale protocol. At $n = 3$ (pilot), signal is visible but ambiguous; the resolution is too low to distinguish real effects from noise. At $n = 30$, the picture sharpens and ambiguous signals resolve. At $n = 300$, fine structure emerges that $n = 30$ cannot see, and some $n = 30$ effects are exposed as small-sample false positives. The three scales are nested (the $n = 30$ seeds are the first 30 of the 300), so where an effect reaches significance at $n = 30$ and vanishes at $n = 300$ we call it a false positive, not a finding at a coarser resolution.

Our contributions are:

1. **A twelve-experiment faultization methodology** that systematically degrades the transformer interface across two phases: perturbation during training (Exp 1-6) and multi-phase morphogenetic interventions (Exp 7-12), to reveal what patterns the system accesses from the latent space.
2. **A four-category classification** (pattern manifestation, pattern fidelity, pattern corruption, free lunch) that organizes what faultization reveals about the system’s relationship to the latent space.
3. **Invariance of final loss across interface architectures.** The same final-loss level is reached through end-to-end backpropagation, through purely local layerwise loss (cell-view, $p = 0.24$ at $n = 30$; $+0.1\%$, $p = 0.53$ at $n = 300$), through chimeric assembly of independently trained networks (all $p > 0.07$), and after a layer is reset to random weights and retrained (all $p > 0.17$ at $n = 30$). The endpoint loss does not depend on the particular interface; whether the underlying solution, representation, or loss-basin is the same is not tested here.
4. **A sharp distinction between an inactive component and a corrupted one.** Frozen (no-further-update) layers are tolerated (-0.1% , $p = 0.41$ at $n = 300$); gradient-negated (adversarial) layers degrade final loss by $+26.3\%$ ($p < 0.0001$, $d = +0.531$ at $n = 300$). A layer that stops learning is absorbed; one driven by inverted gradients is catastrophic. (Frozen layers stay active in the forward pass; the contrast is no-further-plasticity versus adversarial plasticity.)
5. **A three-scale protocol** that treats $n = 3$ pilot findings as coarse signal, $n = 30$ as resolved signal, and $n = 300$ as fine-structure signal, and that here catches a sample-size artifact: head-freezing final-loss significances at $n = 30$ dissolve at $n = 300$.
6. **A dose-controlled null on stress inoculation.** A naive gradual-vs-sudden comparison appears to show that gradual noise builds tolerance, but the gradual schedule delivers only about one-third the cumulative gradient-noise variance of the always-on condition. The dose-matched comparison (gradual vs sudden-half) shows no inoculation ($p = 0.35$ at $n = 30$, $p = 0.20$ at $n = 300$). We report this as a confound caught under dose control rather than a free lunch.
7. **A negative result on rerouting.** The Delayed Gratification Index shows no perturbation response at $n = 30$ or $n = 300$, distinguishing transformer perturbation response from the richer compensatory rerouting observed in biological development.

2. Related Work

2.1 Levin’s Platonic Space Framework

Levin (2026) argues that non-physical patterns, mathematical truths, topological structures, computational regularities, are real, discovered rather than created, and causally relevant to physics, biology, and computer science. Physical systems are interfaces through which patterns from a latent (Platonic) space manifest. The framework proposes that evolution, learning, and engineering exploit these patterns as affordances: once a physical system instantiates the right structure, the

patterns are available for free. The no-free-lunch theorems are constraints on the physical interface; the latent space itself offers enablements that the interface did not pay for.

This framework reframes the question of what a computational system does. Rather than asking “what did the optimizer prescribe?” we ask “what patterns does the system access, and which of those were not paid for by the optimization process?” Faultization, systematic perturbation of the interface, is a natural methodology for probing this question. By degrading the interface, we observe which patterns continue to manifest (pattern fidelity), which are lost (the interface was critical), which are inverted (pattern corruption), and which capabilities appear that were never specified (free lunches). The present paper applies this methodology to a minimal transformer, quantifying the free lunches and mapping the fidelity thresholds.

2.2 Prior Work

Pruning and the lottery ticket hypothesis. Frankle & Carbin (2019) showed that trained networks contain sparse subnetworks (“winning tickets”) that match full-network performance. Subsequent work extended this to structured pruning of attention heads (Michel et al., 2019; Voita et al., 2019). In the Platonic Space framework, pruning simplifies the interface, fewer degrees of freedom may provide cleaner access to the same pattern. Our Experiment 1 freezes heads at *random initialization* rather than after training, so the frozen heads are arbitrary random projections. At $n = 3$, the signal was ambiguous: freezing 8+ heads appeared to improve final loss. At $n = 30$, some final-loss improvements appear significant (freeze 4: $p = 0.001$, freeze 8: $p = 0.023$, freeze 12: $p = 0.016$) alongside a robust mean-trajectory improvement for 4+ frozen heads. At $n = 300$, the final-loss improvements resolve to null (all $p > 0.15$; Spearman $\rho = -0.0045$, $p = 0.84$), but the trajectory improvement strengthens to high significance: freeze 8 ($\Delta = -0.1\%$, $p < 0.0001$, $d = -1.245$), freeze 12 ($\Delta = -0.2\%$, $p < 0.0001$, $d = -1.421$), freeze 16 ($\Delta = -0.2\%$, $p < 0.0001$, $d = -1.312$), confirming that a simpler interface reduces gradient interference and provides cleaner pattern access.

SignSGD and low-precision optimization. Bernstein et al. (2018) established that sign-only gradient updates can match full-precision optimization under appropriate conditions. In pattern-access terms, this asks how much of the gradient signal is essential for the interface to maintain pattern fidelity. At $n = 3$, our Experiment 3 signal was ambiguous: gradient degradation appeared neutral. At $n = 30$, the signal resolved clearly: sign-only gradients significantly degrade performance (+5.0% final loss, $p = 0.002$, $d = +0.614$), and noisy $\sigma = 0.1$ also degrades (+2.4%, $p = 0.032$). The discrepancy between our finding and the SignSGD literature may reflect our model’s small scale and short training duration. At $n = 300$, this finding strengthens: sign-only gradients degrade by +4.9% ($p < 0.0001$, $d = 0.575$), confirming that the discrepancy with SignSGD is robust and not a small-sample artifact.

Noise as regularization. The regularizing effect of gradient noise is well-established (Neelakantan et al., 2015). Our Experiment 3 shows that small noise ($\sigma = 0.01$) produces no significant change ($p = 0.843$), consistent with noise-as-regularization, while large noise ($\sigma = 0.1$) significantly degrades final loss at $n = 30$ (+2.4%, $p = 0.032$) and strongly degrades mean trajectory loss

(+2.4%, $p < 0.001$, $d = +4.306$), indicating a fidelity threshold between $\sigma = 0.01$ and $\sigma = 0.1$, below which the interface maintains clean pattern access, above which it does not.

Local learning rules. Alternatives to end-to-end backpropagation include greedy layerwise pre-training (Bengio et al., 2007), local learning signals (Nokland & Eidnes, 2019), and forward-forward algorithms (Hinton, 2022). Local learning asks whether the final-loss endpoint is invariant to the communication structure of the interface, whether it can be reached through independent channels rather than a single coordinated one. Our cell-view experiment uses local loss (layerwise cross-entropy), eliminating *all* inter-layer gradient flow. The resulting final loss is not significantly different from baseline at both $n = 30$ (-0.9% , $p = 0.237$) and $n = 300$ ($+0.1\%$, $p = 0.53$), with the mean trajectory showing a significant cost ($+0.8\%$, $p < 0.001$, $d = +1.302$). The same endpoint loss is reached through a structurally different interface, though the learning it uses is local only in backward credit assignment: every layer still receives the global next-character target through its own projection head.

Perturbation analysis in neural networks. Ablation studies (Meyes et al., 2019), dropout, and pruning are standard tools, but typically measure *component importance*. We use perturbation to characterize how robustly the system reaches its training endpoint under interface degradation, rather than which parts matter most. This distinction connects our work to Levin’s framework rather than to standard ablation methodology.

What scalar loss leaves undetermined. Several of our comparisons ask whether two training interfaces reach “the same” solution, but every outcome we record is a single scalar, final training loss. Two networks can share a loss value while computing different functions on individual inputs, occupying different regions of weight space, or using different internal representations. The tools built to adjudicate these questions, linear-interpolation and mode-connectivity analysis of the loss landscape (Garipov et al., 2018), representational-similarity measures such as centered kernel alignment (Kornblith et al., 2019), and model stitching between independently trained networks (Bansal et al., 2021), are exactly the measurements we do not make. Throughout, we therefore report equalities of endpoint loss and label them as such; we do not claim shared basins, representations, or functional roles on the strength of a matched scalar.

The morphogenetic framework for algorithms. Zhang, Goldstein, and Levin (2024) applied developmental biology concepts to simple sorting algorithms. Key findings included delayed gratification (temporary performance decrease followed by recovery past pre-damage levels) and fault tolerance that exceeded intact system performance. Our results show the transformer accesses patterns not directly prescribed by the optimizer, most clearly pattern re-binding in complete recovery (Experiment 7), functional role patterns in complete regeneration (Experiment 10), and interface simplification in head-freezing trajectory improvement (Experiment 1). Cell-view (local loss) achieves equivalent final-loss convergence, demonstrating pattern invariance across interface architectures. The Delayed Gratification Index, which Zhang et al. used to detect rerouting, shows no perturbation response at $n = 30$.

Stress inoculation. The phenomenon of gradual stressor exposure building tolerance is well-

documented in biology (Meichenbaum, 1985) and metallurgy (work hardening). In deep learning, curriculum learning (Bengio et al., 2009) and noise scheduling in diffusion models provide partial analogs. Our Experiment 9 set out to test whether gradient-noise exposure shows the same effect. It does not, once total noise dose is controlled: a gradual ramp delivers far less cumulative noise than always-on exposure, and the dose-matched comparison shows no order effect. Section 4.9 reports this as a dose confound rather than a free lunch.

Distributed chess as collective intelligence. Kofman, Campitelli & Levin (2025) extended the morphogenetic framework to chess with autonomous pieces. Our Experiment 6 uses a proper 2×2 factorial design with composite perturbations (each cell has both a forward and gradient perturbation). The key finding is that gradient type dominates: sign-only gradients (+5.2–6.5% final loss) degrade far more than noisy $\sigma = 0.1$ gradients (+2.4–2.6%), regardless of forward perturbation type (sign-only – noisy: -3.8% , $p = 0.022$ at $n = 30$; -2.9% , $p < 0.0001$ at $n = 300$). Their information bottleneck finding (intermediate vision radius outperforms omniscience) does not replicate for attention windowing (no significant final-loss effects at $n = 30$).

3. Methods

3.1 Model Specification

We use a minimal character-level GPT with the following architecture:

Parameter	Value
Layers	4
Embedding dimension	16
Attention heads per layer	4 (16 total)
Head dimension	4
Context length	16
Vocabulary	~28 (a-z + special tokens)
Normalization	RMSNorm
Activation	ReLU
Total parameters	~13,400

The model is implemented in a numpy backend for experiment sweeps. The task is character-level name generation trained on a dataset of ~32k names.

The choice of a minimal model is deliberate and follows Zhang, Goldstein, and Levin’s (2024) rationale for using sorting algorithms: “the benefit of these sorting algorithms is precisely that they are simple, easy to understand, and offer no place for additional complexity to hide.” A minimal interface makes it easier to quantify what patterns ingress and what free lunches appear, complexity cannot hide in an architecture this small.

3.2 Three-Scale Protocol

We adopt an explicit three-scale protocol for interpreting findings. At $n = 3$ (pilot scale), signal is visible but resolution is too low to distinguish real effects from sampling noise; findings at this scale are coarse signal awaiting confirmation. At $n = 30$ (primary scale), paired statistical analysis ($df = 29$) resolves ambiguity, but its power is limited: a two-sided paired t -test at this size reaches 80% power only around $d \approx 0.53$, and has roughly 57% power at $d = 0.4$. Several $d \approx 0.4$ effects that reach significance at $n = 30$ therefore sit in an underpowered band, and, as it turns out, do not survive at $n = 300$. At $n = 300$ (fine-structure scale), effects that are small but real at $n = 30$ either strengthen into clear signals or resolve to null; and new structure that was below the detection threshold at $n = 30$ may emerge.

A finding that is ambiguous at $n = 3$ and resolves clearly at $n = 30$ is not a retraction, it is a finding at the appropriate resolution. A signal that appeared at $n = 3$, resolved to null at $n = 30$, and was confirmed null at $n = 300$ is a signal whose character stabilized as resolution increased; conversely, a signal that was significant at $n = 30$ (sign-only gradient degradation, $p = 0.002$) and strengthened to $p < 0.0001$ at $n = 300$ at a stable effect size is a signal whose precision increased at higher resolution. Higher resolution is not, however, a substitute for experimental control: a confounded comparison (Experiment 9) also grows more significant with n , while remaining uninterpretable. Scale resolves sampling error, not design error.

3.3 Delayed Gratification (DG) Index

As in Zhang, Goldstein, and Levin (2024), we define a metric to detect rerouting behavior, episodes where the system temporarily moves *away* from its goal before recovering past the pre-perturbation level.

Episode detection. We scan the loss trajectory for episodes where: (1) loss increases from a local value L_{start} to a peak L_{peak} , then (2) decreases to a trough L_{trough} below L_{start} . Each such episode has:

- Temporary cost: $C = L_{\text{peak}} - L_{\text{start}}$
- Net gain: $G = L_{\text{start}} - L_{\text{trough}}$

Per-episode DG: $\text{DG}_{\text{episode}} = G/C$

Aggregate DG Index: The mean DG across all detected episodes in a training run.

At $n = 3$, the DG Index appeared to scale with perturbation severity. At $n = 30$, this signal resolved to null: no perturbation condition produces a statistically significant DG increase ($p > 0.19$ for all). DG captures a real stochastic property of loss trajectories but does not function as a perturbation response measure at $n = 30$. At $n = 300$, the DG null holds: the metric still does not track perturbation, confirming that DG captures stochastic loss-trajectory structure rather than perturbation response.

3.4 Experiment 1: Head Freezing (Interface Simplification)

Motivation: the frozen-cell perturbation of Zhang, Goldstein, and Levin (2024). We simplify the interface by freezing randomly selected attention heads at initialization values throughout training, reducing the degrees of freedom through which optimization patterns manifest.

Protocol: Sweep over $\{0, 1, 2, 4, 8, 12, 16\}$ frozen heads. Frozen heads participate in the forward pass but receive no gradient updates.

3.5 Experiment 2: Cell-View GPT (Pattern Invariance)

Motivation: the cell-view sorting algorithms of Zhang, Goldstein, and Levin (2024). Each transformer layer is treated as an autonomous agent. This tests whether the optimization pattern is invariant to the communication architecture of the interface, whether it can be accessed through independent channels.

Protocol: Local loss (layerwise cross-entropy) at all layer boundaries. Each layer receives only its own local loss signal, computed as a cross-entropy loss from a per-layer projection head. No inter-layer gradient flow.

3.6 Experiment 3: Gradient Degradation (Pattern Fidelity)

Motivation: the noisy signaling channels of Zhang, Goldstein, and Levin (2024). We degrade the gradient signal to probe the fidelity threshold, how much interface corruption the pattern tolerates before access is lost. Four corruption methods:

Method	Description
Noisy ($\sigma = 0.01$)	Additive Gaussian noise, small scale
Sign-only	Gradient reduced to $\{-1, 0, +1\}$, magnitude discarded
Quantized (3-level)	Gradient rounded to three levels (ternary: $-$, 0 , $+$ at a fixed scale)
Noisy ($\sigma = 0.1$)	Additive Gaussian noise, large scale

3.7 Training Protocol and Statistical Methods

All experiments use: 200 training steps, 30 independent runs per condition (seeds 42–71), loss and per-head metrics recorded at every step. The 200-step horizon captures early learning dynamics. The $n = 300$ replication re-runs every experiment under the identical deterministic harness with 300 matched seeds (42 to 341) and the same 200-step horizon; unless a figure is labelled $n = 30$, the $n = 300$ values quoted in Section 4 come from that committed re-analysis. Because the $n = 30$ seeds are the first 30 of the $n = 300$ block rather than an independent set, the two scales are nested, and an effect that reaches significance at $n = 30$ and dissolves at $n = 300$ is most honestly read as a small-sample false positive rather than a phenomenon that was real at lower resolution.

Statistical analysis. All comparisons use two-sided paired t -tests, with runs matched by seed across conditions ($n = 30$, $df = 29$). Pairing by seed controls for initialization variance. We report effects as statistically significant at $p < 0.05$ and marginal at $0.05 < p < 0.10$. With 30 paired observations power is limited: 80% power arrives only near $d \approx 0.53$, and detection at $d = 0.4$ is about 57%, so p -values in the 0.01 to 0.05 range at this scale are provisional until confirmed at $n = 300$. Effect sizes are reported as Cohen’s d for paired differences. We distinguish *statistically supported* findings ($p < 0.05$) from *observational* patterns, and we try not to conflate a shrinking p -value under larger n with a growing effect: where an effect size falls while p drops (as in Experiment 9), we say so.

A note on equivalence. Many of our comparisons are null-like: invariance, recovery, no-transplant-advantage, no cost from freezing. A non-significant two-sided test ($p > 0.05$) does not by itself establish equivalence; it means the data did not provide evidence against an exact-zero difference. A proper equivalence claim would predefine a smallest effect of interest and run two one-sided tests, or TOST (Lakens et al., 2018). We did not run such tests, so we describe these results as “no detectable difference in final loss” and treat “equivalent” as shorthand for that, not as a demonstrated equivalence.

Multiple comparisons. We do not apply formal multiple-comparison correction across the twelve experiments, on the view that each tests a distinct perturbation type with its own pre-specified primary comparison. This does not cover the multiplicity *within* an experiment: head freezing sweeps seven levels, vision radius five windows, and chimeras four assemblies, and there we screen several contrasts at once. Marginal results inside these families ($0.01 < p < 0.05$), including the $n = 300$ vision-radius effects (window 1 at $p = 0.021$, window 8 at $p = 0.022$), would not survive a Holm correction across their family, which would require the smallest p below about 0.01; we treat them as exploratory rather than confirmatory. The strongest findings ($p < 0.0001$: competing objectives, sign-only degradation, and the cell-view trajectory cost) survive any reasonable correction.

Confirmatory vs. exploratory. We distinguish three categories: pre-specified primary comparisons (each experiment’s main effect), secondary metrics (mean trajectory loss, DG index), and fine-structure analyses (layer-specific or dose-response patterns discovered at $n = 300$). Only primary comparisons are treated as confirmatory; secondary and fine-structure findings are explicitly exploratory.

3.8 Experiment 4: Vision Radius Sweep (Pattern Visibility)

Motivation: Kofman, Campitelli & Levin’s (2025) vision radius experiment in distributed chess. We restrict each attention head’s context window, probing the information geometry of pattern access, how much of the input the interface needs to see for the pattern to manifest.

Protocol: Sweep over window sizes $\{1, 2, 4, 8, 16\}$ plus unmodified baseline. Window=16 equals block size (sanity check).

3.9 Experiment 5: Communication Topology (Layerwise Autonomy)

Motivation: The chess paper’s relay chains. We scale the fraction of gradient signal passed through layer boundaries, testing whether the pattern is accessible through independent channels or requires coordinated inter-layer communication.

Protocol: Five topologies parameterized by gradient pass fraction:

Topology	Pass Fraction	Description
Full	1.00	Standard backpropagation (baseline)
Heavy	0.75	75% of gradient signal passes through
Half	0.50	50% of gradient signal passes through
Light	0.25	25% of gradient signal passes through
Cell-view	0.00	No inter-layer gradient flow

3.10 Experiment 6: Courage vs. Caution (Channel Sensitivity)

Motivation: Kofman et al.’s “cautious position, courageous moves” strategy. We create a 2×2 factorial design with composite perturbations, each cell applies *both* a forward perturbation and a gradient perturbation simultaneously, to probe which interface channel is the critical signal for pattern access:

	Cautious Gradients (sign-only)	Courageous Gradients (noisy $\sigma = 0.1$)
Cautious Forward (tiny noise $\sigma = 0.001$)	(a) Tiny noise + sign-only	(b) Tiny noise + noisy $\sigma = 0.1$
Courageous Forward (dropout $p = 0.1$)	(c) Dropout + sign-only	(d) Dropout + noisy $\sigma = 0.1$

3.11 Experiment 7: Recovery After Damage (Pattern Re-Binding)

Motivation: Levin’s regeneration paradigm, does a damaged organism recover after the damage is removed? Does the pattern re-bind to the interface after disruption?

Protocol: Three-phase training with matched controls. - Phase 1: Normal training (200 steps) - Phase 2: Damage, freeze 8 random heads (100 steps) - Phase 3: Recovery, unfreeze all heads, continue (200 steps) - Control: Undamaged training for the same total duration (500 steps)

Learning rate schedule is matched across phases and control ($\text{lr}(t) = \text{lr}_0 \cdot (1 - t/500)$). Paired by seed ($n = 30$).

3.12 Experiment 8: Chimera Assembly (Basin Universality)

Motivation: Chimeric organisms assembled from parts of different embryos. Can a network assembled from independently-trained components retrain to the same final loss? This asks whether any layer assignment prevents the assembled model from reaching the control loss level.

Protocol: Train model A (seeds 42–71) and model B (seeds 1042–1071) independently for 200 steps. Assemble four chimera types by selecting each layer from either model:

Chimera	Layer 0	Layer 1	Layer 2	Layer 3
AABB	A	A	B	B
ABAB	A	B	A	B
BBAA	B	B	A	A
ABBA	A	B	B	A

Shared parameters (embeddings, output projection) come from model A. Continue training each chimera for 200 more steps. Control: model A continues training without modification.

3.13 Experiment 9: Gradual vs. Sudden Damage (Dose vs. Exposure History)

Motivation: Biological stress inoculation, gradual exposure to a stressor builds tolerance that sudden exposure does not. In pattern-access terms: does the history of interface degradation, holding total degradation fixed, change which patterns become accessible? Isolating exposure history requires controlling for total noise dose, which the conditions below let us do.

Protocol: Four conditions, all 200 steps: - Control: no gradient noise - Sudden full: noisy gradients ($\sigma = 0.1$) for all 200 steps - Gradual: linear ramp from $\sigma = 0$ to $\sigma = 0.1$ over 200 steps - Sudden half: no noise for first 100 steps, then $\sigma = 0.1$ for remaining 100 steps

The gradual condition reaches the same peak noise level as sudden full but arrives there incrementally. This means the four conditions do not deliver the same total noise. Cumulative injected gradient-noise variance, $\sum_t \sigma(t)^2$, is 2.0 for sudden full (200 steps at $\sigma^2 = 0.01$), 1.0 for sudden half (100 steps), and approximately 0.67 for the gradual ramp ($\sum_t (0.1 \cdot t/200)^2 \approx 0.01 \cdot 200/3$). The gradual condition therefore receives about one-third the dose of sudden full and two-thirds the dose of sudden half. Because dose and exposure history are confounded by this design, the gradual-vs-sudden-full contrast cannot isolate an inoculation effect. No pair here is exactly dose-matched, but the closest is gradual (dose 0.67) versus sudden half (dose 1.0), and it has the gradual condition receiving *less* total noise, so a genuine inoculation advantage for gradual would have to show up despite a dose handicap. We report both contrasts, and treat “dose-matched” as shorthand for this nearest-available comparison rather than an exact match.

3.14 Experiment 10: Regeneration (Functional Role Patterns)

Motivation: Regeneration after tissue destruction. Can a trained network rebuild a destroyed layer? In the Platonic Space framework, this tests whether the pattern determines the functional role of a position, regardless of what substrate occupies it.

Protocol: Two-phase training. - Phase 1: Normal training (200 steps) - Phase 2: Reset one layer's weights to random initialization (new random seed). Zero the corresponding Adam optimizer state. Continue training for 200 more steps. - Control: no reset, continue training. - Test all 4 layers independently. Paired by seed ($n = 30$).

3.15 Experiment 11: Transplantation (Context-Dependent Roles)

Motivation: Organ transplantation, is a layer from a separately-trained donor network accepted better than a random replacement? This asks whether a donor layer's learned weights give any measurable head start over a random replacement at the same position.

Protocol: Train model A and model B independently (200 steps each, different seeds). For each layer L : - Transplant: replace layer L of model A with layer L from model B, continue training 200 steps - Random reset: replace layer L of model A with random weights, continue training 200 steps - Control: model A continues without modification

Adam buffers are zeroed for the replaced layer in both transplant and random conditions. Paired by seed ($n = 30$).

3.16 Experiment 12: Competing Objectives (Pattern Corruption)

Motivation: Inter-organ conflict, what happens when part of the network fights the objective? This distinguishes between pattern unavailability (the interface is silent) and pattern corruption (the interface actively inverts the signal).

Protocol: Two-phase training. - Phase 1: Normal training (200 steps) - Phase 2: Negate gradients for layers 2-3 while layers 0-1 train normally (200 steps). The negated layers receive gradient signals that push them *away* from the loss minimum. - Comparison: freeze layers 2-3 (zero their gradients) instead of negating - Control: normal training for 400 steps total

This tests whether layers 0-1 can compensate for actively adversarial downstream layers (negation) versus merely inactive ones (freezing).

4. Results

4.1 Experiment 1: Head Freezing: Interface Simplification

At $n = 3$: The signal was ambiguous. Head freezing appeared to improve final loss, freezing 8+ heads seemed to produce statistically significant improvement ($p = 0.009$ in pilot data). The coarse resolution could not distinguish this from noise.

At $n = 30$: At this scale, some final-loss improvements appear significant: freeze 2 (-0.3% , $p = 0.025$, $d = -0.430$), freeze 4 (-0.5% , $p = 0.001$, $d = -0.647$), freeze 8 (-0.5% , $p = 0.023$, $d = -0.438$), freeze 12 (-0.7% , $p = 0.016$, $d = -0.467$). Freeze 1 ($p = 0.253$) and freeze 16 ($p = 0.322$) are non-significant. The overall monotonic trend is weak (Spearman $\rho = -0.015$, $p = 0.83$). Mean trajectory loss shows a robust improvement for 4+ frozen heads, with larger effect sizes than final loss. This is a different, finer signal that became visible at higher resolution.

At $n = 300$: The final-loss improvements resolve to null (all $p > 0.15$; Spearman $\rho = -0.0045$, $p = 0.84$). The $n = 30$ final-loss significances were artifacts of moderate sample size. The trajectory improvement strengthens to high significance: freeze 4 mean-loss $\Delta = -0.1\%$ ($p < 0.0001$, $d = -0.971$), freeze 8 $\Delta = -0.1\%$ ($p < 0.0001$, $d = -1.245$), freeze 12 $\Delta = -0.2\%$ ($p < 0.0001$, $d = -1.421$), freeze 16 $\Delta = -0.2\%$ ($p < 0.0001$, $d = -1.312$). The monotonic dose-response in trajectory improvement, combined with no final-loss cost at $n = 300$, confirms that frozen random-projection heads reduce gradient interference throughout training without affecting convergence.

Table 1. Head freezing results (means $\pm$ std across 30 runs).

Frozen Heads	Final Loss	Mean Loss	DG Index
0 (baseline)	2.557 ± 0.407	2.627 ± 0.028	0.680 ± 1.131
1	2.553 ± 0.406	2.627 ± 0.028	0.632
2	2.549 ± 0.405	2.626 ± 0.028	0.689
4	2.543 ± 0.397	2.625 ± 0.028	0.636
8	2.544 ± 0.396	2.623 ± 0.028	0.640
12	2.539 ± 0.400	2.622 ± 0.028	0.614
16	2.547 ± 0.407	2.620 ± 0.028	0.570

Final loss shows improvements at several freezing levels: freeze 2 ($p = 0.025$, $d = -0.430$), freeze 4 ($p = 0.001$, $d = -0.647$), freeze 8 ($p = 0.023$, $d = -0.438$), freeze 12 ($p = 0.016$, $d = -0.467$). However, these resolve to null at $n = 300$ (all $p > 0.15$). Mean trajectory loss shows a robust improvement for 4+ frozen heads: freeze 4 ($p < 0.001$, $d = -1.008$), freeze 8 ($p < 0.001$, $d = -1.228$), freeze 12 ($p < 0.001$, $d = -1.366$), freeze 16 ($p < 0.001$, $d = -1.070$). The effect size is 0.1–0.3% of mean loss, statistically robust but practically small.

A plausible reading is that fewer trainable degrees of freedom reduce destructive gradient interactions, letting the remaining parameters navigate the loss landscape more efficiently. What we can state without interpretation is narrower: the final-loss endpoint is unchanged, and the mean training trajectory improves by 0.1 to 0.3%. The effect is real but small, and it lives in the trajectory, not the endpoint.

The DG Index does not increase with freezing. No freezing level produces a significant DG change.

Free lunch: Trajectory improvement without paying for it. The optimizer specifies loss minimization; it does not specify that removing parameters should improve the trajectory. The improvement

is a free lunch, a pattern the system accesses from the latent space that was not prescribed by the optimization objective.

Classification: Final loss unchanged under interface simplification, with a small trajectory improvement (an informal free lunch in the sense of Section 1).

4.2 Experiment 2: Cell-View GPT: Pattern Invariance

At $n = 3$: The signal was ambiguous. Cell-view appeared to elevate DG substantially (+25.5%), suggesting possible rerouting behavior.

At $n = 30$: The DG signal resolved to null ($p = 0.61$). The final-loss signal is non-significant: cell-view produces near-identical final loss to baseline (-0.9% , $p = 0.237$, $d = -0.220$). The mean trajectory loss shows a significant cost ($+0.8\%$, $p < 0.001$, $d = +1.302$).

At $n = 300$: The final-loss result holds at high power: cell-view final loss $+0.1\%$ ($p = 0.53$, $d = +0.036$), no detectable difference from baseline. Mean trajectory: $+0.2\%$ ($p < 0.0001$, $d = +0.731$). DG: $p = 0.14$ (ns). Local layerwise optimization reaches the same final loss as end-to-end backpropagation, with a small trajectory cost. Whether it reaches the same solution or loss-basin is not something a scalar loss can tell us.

Table 2. Cell-view (local loss) vs. baseline (means $\pm$ std across 30 runs).

Condition	Mean Loss	Final Loss	DG Index
Baseline	2.627 ± 0.028	2.557 ± 0.407	0.680
Cell-view	2.647 ± 0.025	2.535 ± 0.384	0.568

Replacing end-to-end backpropagation with local loss produces no detectable difference in final loss (-0.9% , $p = 0.237$, $d = -0.220$) and a significant mean-trajectory cost ($+0.8\%$, $p < 0.001$, $d = +1.302$). The same final-loss endpoint is reached via a structurally different interface: independent layerwise channels rather than a single coordinated gradient flow.

We read this as invariance of the endpoint, not of the underlying solution: a scalar loss cannot say whether the two interfaces settle into the same region of weight space. And the local rule is local only in backward credit assignment, not in supervision. Each layer’s projection head is still trained against the global next-character target, so the “global convergence from local rules” is really convergence from locally-routed gradients under globally-supplied labels, at the cost of extra auxiliary heads.

Free lunch: In the informal sense of Section 1, the same final loss is reached without end-to-end gradient information. It is not free of resources: the layerwise projection heads are added parameters and supervision the end-to-end model does not carry.

Classification: Final-loss invariance across interface architectures.

4.3 Experiment 3: Gradient Degradation: Pattern Fidelity

At $n = 3$: The signal was ambiguous. All four gradient degradation methods appeared neutral ($p > 0.26$), and small noise appeared to help.

At $n = 30$: The ambiguous signal resolved. Three of four methods significantly degrade final loss; one is genuinely tolerated.

At $n = 300$: The threshold between fidelity and loss of access sharpens. Noise at $\sigma = 0.01$ remains non-significant (+0.0%, $p = 0.86$), confirming genuine pattern fidelity, the pattern manifests cleanly through this level of interface degradation. All three degradation conditions strengthen: noise at $\sigma = 0.1$ (+2.3%, $p < 0.0001$, $d = 0.391$), sign-only (+4.9%, $p < 0.0001$, $d = 0.575$), and quantized 3-level (+3.6%, $p < 0.0001$, $d = 0.529$). The sign-only effect strengthened from $p = 0.002$ at $n = 30$ to $p < 0.0001$ at $n = 300$.

Table 3. Gradient degradation results (means across 30 runs).

Method	Final Loss	$\Delta\%$	p (final)	Mean Loss	p (mean)	d (mean)
Baseline	2.557	,	,	2.627	,	,
Noisy ($\sigma = 0.01$)	2.560	+0.1%	0.843	2.628	0.333	+0.180
Noisy ($\sigma = 0.1$)	2.618	+2.4%	0.032*	2.690	<0.001***	+4.306
Sign-only	2.685	+5.0%	0.002**	2.729	<0.001***	+5.696
Quantized (3-level)	2.653	+3.8%	0.008**	2.693	<0.001***	+4.457

Three of four methods significantly degrade final loss: sign-only (+5.0%, $p = 0.002$, $d = +0.614$), quantized (+3.8%, $p = 0.008$, $d = +0.519$), and noisy $\sigma = 0.1$ (+2.4%, $p = 0.032$, $d = +0.411$). Noisy $\sigma = 0.1$ also strongly degrades mean trajectory (+2.4%, $p < 0.001$, $d = +4.306$). Only small noise ($\sigma = 0.01$) is genuinely tolerated ($p = 0.843$ final, $p = 0.333$ mean). Mean loss effects are highly significant for sign-only and quantized ($p < 0.001$) with large Cohen’s d values (4.5–5.7). The “noise helps” effect (noisy $\sigma = 0.01$ improving loss) is not supported ($p = 0.333$ for mean loss).

There is a threshold between $\sigma = 0.01$ and $\sigma = 0.1$: below it, gradient noise leaves final loss unchanged; above it, it degrades. The sign-only result needs care, and the paper earlier stated it backwards. Replacing the gradient with its sign *keeps* direction and *discards* magnitude, so the +5.0% degradation shows that magnitude information matters here, that keeping only the sign is not enough. It does not show that sign information was destroyed (sign-only preserves the sign), and it is not a controlled comparison with signSGD, which tunes its own step size and, in its momentum variant, was the configuration reported to approach Adam. What the gradient experiments do support is narrower: a threshold on how much gradient corruption the endpoint

tolerates, and, read alongside Experiment 6, that corrupting the gradient channel costs more than corrupting the forward channel.

Classification: Final loss is tolerant of small gradient noise ($\sigma = 0.01$) and degrades above a threshold; discarding gradient magnitude (sign-only) costs the most.

4.4 Experiment 4: Vision Radius Sweep: Pattern Visibility

At $n = 3$: The signal was ambiguous. An information bottleneck effect appeared possible, intermediate window sizes seemed to outperform full context.

At $n = 30$: The ambiguous signal resolved to null for final loss across all window sizes. Tiny mean-trajectory effects are statistically detectable but not practically meaningful.

At $n = 300$: Small effects emerge that were invisible at $n = 30$. Window 1 harms performance (+0.3%, $p = 0.021$, $d = +0.134$), windows 2 and 4 remain non-significant ($p = 0.93$ and $p = 0.93$ respectively), window 8 shows a small improvement (−0.1%, $p = 0.022$, $d = −0.133$), and window 16 is identical to baseline. These per-window p -values (0.021, 0.022) do not survive correction across the five windows and are exploratory. Taken together they are suggestive of a weak information-bottleneck effect below the $n = 30$ detection threshold, and no more than suggestive.

Table 4. Vision radius results (means across 30 runs).

Window	Final Loss	p (final)	Mean Loss	p (mean)	d (mean)
Baseline (full)	2.557	,	2.627	,	,
1	2.562	0.618	2.639	<0.001***	+1.847
2	2.555	0.898	2.628	0.156	,
4	2.549	0.443	2.625	<0.001***	-0.735
8	2.553	0.304	2.626	0.037*	-0.400
16	2.557	1.000	2.627	1.000	0.00

No window size significantly changes final loss at $n = 30$ ($p > 0.30$ for all). Window=16 reproduces the baseline (sanity check). Mean-loss effects exist but are negligibly small. The chess paper’s finding that intermediate vision radius outperforms omniscience does not produce meaningful effects for attention windowing at this scale.

Final loss is nearly flat across window sizes: the endpoint is reached regardless of how much context each attention head sees, and only extreme restriction (window 1) begins to harm. On character-level name generation this most likely reflects how little long-range context the task demands, as much as any property of the architecture; a task built around long-range dependencies would be needed to separate the two.

Classification: Pattern fidelity, the pattern manifests through the interface at all tested visibility levels.

4.5 Experiment 5: Communication Topology: Layerwise Autonomy

At $n = 3$: The signal was ambiguous. A U-shaped loss curve appeared possible, partial communication seemed to outperform both full and no communication.

At $n = 30$: The U-shaped curve resolved to flat across all gradient fractions, including zero communication (cell-view). Partial gradient flow is absorbed without meaningful degradation.

At $n = 300$: The architecture’s indifference to gradient fraction holds at high power. Heavy ($p = 0.39$), half ($p = 0.18$), and light ($p = 1.00$) communication topologies remain non-significant. Cell-view final loss is non-significant ($p = 0.53$), consistent with Experiment 2’s finding that local loss achieves equivalent convergence. Only cell-view mean trajectory shows significant cost ($+0.8\%$, $p < 0.0001$, $d = +0.731$).

Table 5. Communication topology results (means across 30 runs).

Topology	Fraction	Final Loss	p (final)	Mean Loss	p (mean)
Full	1.00	2.557	,	2.627	,
Heavy	0.75	2.558	0.379	2.627	0.314
Half	0.50	2.555	0.589	2.627	0.187
Light	0.25	2.557	0.915	2.627	0.895
Cell-view	0.00	2.535	0.237	2.647	$<0.001^{***}$

Partial gradient flow (25–75%) produces no meaningful degradation. Heavy (75%), half (50%), and light (25%) are all non-significant for final loss ($p > 0.37$). Cell-view (0%) shows no detectable difference in final loss (-0.9% , $p = 0.237$, $d = -0.220$) but significantly elevated mean trajectory loss ($+0.8\%$, $p < 0.001$, $d = +1.302$).

This flat response is hard to read as tolerance. Scaling a gradient by a positive constant is close to a no-op under Adam, which normalizes by a running estimate of the gradient’s second moment and was designed to be invariant to diagonal rescaling (Kingma & Ba, 2015). The indifference at fractions 0.25 to 0.75 may therefore reflect the optimizer absorbing the manipulation rather than any autonomy of the layers; a plain-SGD or momentum-SGD control, which we did not run, would be needed to separate the two. The zero-flow (cell-view) condition is different in kind, since it replaces inter-layer gradients with local losses rather than scaling them, and it too reaches the same final loss with a trajectory cost.

Classification: Final loss is unchanged across gradient-pass fractions and at zero inter-layer communication, though the partial-flow result is confounded with Adam’s rescaling invariance.

4.6 Experiment 6: Courage vs. Caution: Channel Sensitivity

At $n = 3$: The signal was ambiguous. The courage/caution matrix appeared to produce inconsistent results without clear pattern.

At $n = 30$: The 2×2 factorial design with composite perturbations reveals a clear pattern: gradient type dominates over forward perturbation type.

At $n = 300$: The gradient-type dominance is confirmed at high statistical power. All four composite conditions significantly degrade final loss: cautious/cautious (+5.0%, $p < 0.0001$, $d = +0.572$), cautious/courageous (+2.3%, $p < 0.0001$, $d = +0.386$), courageous/cautious (+5.2%, $p < 0.0001$, $d = +0.549$), courageous/courageous (+2.3%, $p < 0.0001$, $d = +0.390$). The sign-only vs. noisy gradient contrast: -2.8% ($p < 0.0001$, $d = -0.297$). These $n=300$ results confirm the pattern already significant at $n=30$.

Table 6. Courage vs. caution results, 2×2 factorial with composite perturbations (means across 30 runs).

Condition	Final Loss	p (final)	Mean Loss	p (mean)	d (mean)
Baseline	2.557	,	2.627	,	,
(a) Caut./Caut. (tiny noise + sign-only)	2.689	0.002**	2.729	<0.001***	+5.935
(b) Caut./Cour. (tiny noise + noisy $\sigma=0.1$)	2.619	0.032*	2.690	<0.001***	+4.309
(c) Cour./Caut. (dropout + sign-only)	2.722	0.001***	2.730	<0.001***	+5.155
(d) Cour./Cour. (dropout + noisy $\sigma=0.1$)	2.622	0.035*	2.695	<0.001***	+4.479

The key finding is that **gradient type dominates**: conditions with sign-only gradients (a, c) degrade final loss by +5.2–6.5%, while conditions with noisy $\sigma = 0.1$ gradients (b, d) degrade by +2.4–2.6%. All four conditions significantly degrade final loss ($p < 0.04$), but sign-only conditions show roughly double the effect. The forward perturbation type (tiny noise vs. dropout) has a smaller effect. The sign-only vs. noisy gradient contrast (averaging across forward types) yields -3.8% final loss ($p = 0.022$, $d = -0.440$ at $n = 30$; -2.8% , $p < 0.0001$, $d = -0.297$ at $n = 300$) and -1.5% mean loss ($p < 0.001$, $d = -1.921$). All conditions also strongly degrade mean

trajectory loss.

The interface has an asymmetric channel sensitivity: degrading the gradient (whether by sign-only or by noise) costs more than degrading the forward signal (dropout or tiny noise). The interface is not uniformly fragile; it has a sensitive channel (gradients) and a robust one (forward activations). Which specific property of the gradient matters most is left open by this design, since the sign-only and noisy conditions differ in more than one way and share a fixed Adam step size.

Classification: Channel sensitivity: the gradient channel is more sensitive than the forward channel; among gradient corruptions, discarding magnitude (sign-only) costs most.

4.7 Experiment 7: Recovery After Damage: Pattern Re-Binding

At $n = 3$: The signal was ambiguous. Recovery appeared complete but with too few observations to confirm.

At $n = 30$: Complete recovery is unambiguously confirmed across all 30 runs.

At $n = 300$: A tiny but statistically significant residual emerges at high power. Recovery vs. control: $+0.1\%$ ($p = 0.030$, $d = +0.126$). Final-loss ratio: 1.0009 ± 0.0072 . 272 of 300 runs recovered, with mean overshoot -0.0009 ± 0.0017 . Recovery time: 0.8 ± 1.2 steps. The effect is real but negligibly small, the residual is within practical equivalence.

Table 7. Recovery after transient damage ($n = 30$, 500 total steps).

Metric	Recovery	Control	p (paired)
Final loss	2.451 ± 0.370	2.451 ± 0.375	0.886
Final ratio (rec/ctrl)	1.0000 ± 0.008	,	,
Recovery time	1 ± 1 steps	,	30/30 recovered
Overshoot	-0.0014 ± 0.0015	,	,

The damaged-then-recovered model reaches the same final loss as the undamaged control ($p = 0.886$, ratio 1.0000). All 30 runs recovered within a mean of 1 step after damage removal. No meaningful overshoot was observed (mean overshoot = -0.0014). The 100 steps of training with 8 frozen heads had no lasting effect, the “damage” was entirely absorbed by subsequent normal training.

Once the freeze is lifted, the model returns to the control final loss. The “damage” here is mild, though: freezing 8 heads is the Experiment 1 perturbation, which on its own leaves the endpoint unchanged and slightly improves the trajectory. So this is better described as removal of a benign constraint than as regeneration after injury, and the near-instant return (about 1 step) is consistent with there being little to recover from. What the result does show is that 100 steps under this constraint leave no lasting mark on the final loss. At $n = 300$, a tiny residual becomes detectable ($p = 0.030$, $d = +0.126$) but is within any practical tolerance.

Free lunch: In the informal sense of Section 1, the endpoint is reached again without a dedicated recovery mechanism. It is not free of cost: it takes 200 further training steps, and the constraint recovered from was close to harmless to begin with.

Classification: Final loss returns to the control level after a transient, mild constraint is lifted.

4.8 Experiment 8: Chimera Assembly: Basin Universality

At $n = 3$: The signal was ambiguous. Chimeras appeared to converge but from very few observations.

At $n = 30$: Convergence is confirmed for all chimera types. The specific layer assignment (which layers come from which model) does not matter.

At $n = 300$: All chimera types remain non-significant (AABB $p = 0.35$, ABAB $p = 0.12$, BBAA $p = 0.079$, ABBA $p = 0.31$), with all conditions falling within 0.3% of control ($n = 300$ ctrl = 2.4480). BBAA shows a marginal trend ($p = 0.079$) but does not reach significance. All chimera configurations reach final loss within 0.3% of control.

Table 8. Chimera assembly results ($n = 30, 200 + 200$ steps).

Condition	Initial Loss	Final Loss	vs Control p
Control (A continues)	,	2.494 ± 0.344	,
AABB	2.569	2.484	0.265
ABAB	2.563	2.494	0.985
BBAA	2.605	2.472	0.076+
ABBA	2.542	2.487	0.559

All chimera types reach the same final loss as the control ($p > 0.07$ for all). Despite starting at elevated loss (2.54–2.61 vs. control ~ 2.49), the chimeras retrain to the control final-loss level.

A chimera assembled from parts of two independently-trained models, after 200 further training steps, reaches the same final loss as an unmodified continuation. This is a real and somewhat surprising result about the endpoint. It is not evidence of basin universality: whether the retrained chimera occupies the same loss-basin as either parent, or a third one, would require interpolation or stitching analysis (Bansal et al., 2021) we did not run. What the data support is that no chimeric assignment prevents retraining from reaching the control loss level, most likely because 200 steps is enough for the shared embeddings and output projection to re-fit whatever the swapped layers provide.

Classification: Chimeras reach control-level final loss (no significant difference); an equality of endpoint loss, not a demonstrated shared basin.

4.9 Experiment 9: Gradual vs. Sudden Damage: A Dose Confound, Not Inoculation

At $n = 3$: The signal was ambiguous. The gradual vs. sudden comparison appeared promising but underpowered.

At $n = 30$: A naive reading looks like stress inoculation. Sudden full noise degrades final loss by +2.4% vs. control ($p = 0.032$, $d = +0.411$), while the gradual ramp to the same peak shows no significant degradation (+0.3%, $p = 0.64$). The direct gradual-vs-sudden-full contrast is significant (−2.1%, $p = 0.032$, $d = -0.412$). But sudden half (the dose-matched comparison condition) degrades by +1.0% ($p = 0.040$, $d = +0.393$), and the dose-matched contrast, gradual vs. sudden half, is not significant (−0.7%, $p = 0.35$, $d = -0.172$). The apparent inoculation tracks how much noise each condition received, not the order in which it arrived.

At $n = 300$: The dose pattern sharpens and the inoculation reading does not survive. Degradation is monotone in cumulative noise dose: sudden full (dose 2.0) +2.3% ($p < 0.0001$, $d = +0.391$), sudden half (dose 1.0) +0.3% ($p = 0.19$, ns, $d = +0.077$), gradual (dose 0.67) +0.5% ($p = 0.008$, $d = +0.155$). The gradual-vs-sudden-full contrast is large and highly significant (−1.7%, $p < 0.0001$, $d = -0.374$), but it confounds a 3-to-1 difference in total noise with the difference in exposure history. The dose-matched contrast, gradual vs. sudden half, is not significant (+0.3%, $p = 0.20$, $d = +0.075$) and points the opposite direction from inoculation: at matched dose, gradual is if anything slightly worse, not better, than sudden. The gradual mean trajectory sits marginally below control (−0.1%, $p < 0.0001$, $d = -0.469$) because, over a 0-to-0.1 ramp, the network spends most of training at near-zero noise, which is the noise-as-regularization regime of Experiment 3, not evidence of acquired tolerance to a stressor.

Table 9. Gradual vs. sudden noise ($n = 30$, 200 steps). Cumulative dose is $\sum_t \sigma(t)^2$.

Condition	Dose	Final Loss	p (vs ctrl)	Mean Loss	p (mean)
Control	0.0	2.557 ± 0.407		2.627 ± 0.028	
Sudden full ($\sigma = 0.1$)	2.0	2.618 ± 0.388	0.032*	2.690 ± 0.028	<0.001***
Sudden half (step 100)	1.0	2.582 ± 0.416	0.040*	2.630 ± 0.028	0.004**
Gradual (0 to 0.1)	0.67	2.564 ± 0.408	0.641	2.622 ± 0.027	<0.001***

What the data support. Final-loss degradation rises with total injected noise dose across all three perturbed conditions, at both scales. The gradual condition is not protected by its schedule; it is simply exposed to less noise. When dose is held fixed (gradual vs. sudden half), the difference between gradual and sudden exposure is statistically indistinguishable from zero ($p = 0.35$ at $n = 30$, $p = 0.20$ at $n = 300$). We therefore do not claim a stress-inoculation effect, and we do not treat this experiment as a free lunch. The only robust statement is the dose-response one:

more cumulative gradient noise produces more final-loss degradation, consistent with the fidelity threshold of Experiment 3.

Classification: Pattern fidelity (dose-dependent degradation). No inoculation effect under dose control; the apparent gradual-vs-sudden advantage is a confound between exposure history and total noise dose.

4.10 Experiment 10: Regeneration: Functional Role Patterns

At $n = 3$: The signal was ambiguous. Regeneration appeared possible but with too few observations to confirm the completeness.

At $n = 30$: Complete regeneration is confirmed for all four layers.

At $n = 300$: Fine structure emerges in layer-specific regeneration completeness. All four layers show small but significant residual deficits: Layer 0 (+0.3%, $p = 0.003$, $d = +0.173$), Layer 1 (+0.2%, $p = 0.007$, $d = +0.157$), Layer 2 (+0.1%, $p = 0.024$, $d = +0.131$), Layer 3 (+0.1%, $p = 0.037$, $d = +0.121$). Completeness: L1 0.988, L2 0.994, L3 1.021 (L0 has a data issue; use final loss comparison only). The overall confidence interval tightens substantially relative to $n = 30$, revealing that regeneration is near-complete but not perfectly uniform across layers. The residuals are tiny and graded by layer position.

Table 10. Layer regeneration after random reset ($n = 30, 200 + 200$ steps).

Reset Layer	Immediate Damage	Final Loss	Completeness	vs Control p
Control (no reset)	,	2.457 ± 0.341	,	,
Layer 0	-0.136	2.467	0.998	0.174
Layer 1	-0.152	2.459	1.046	0.550
Layer 2	-0.157	2.456	1.015	0.808
Layer 3	-0.146	2.457	0.981	0.977

Immediate Damage in Table 10 is (post-reset loss — end-of-phase-1 loss), evaluated one step after the reset. The values are small and, for these layers, slightly negative; this is an artifact of comparing a fresh forward pass against the last training minibatch, not a sign that destroying a layer lowers loss (a random-weight layer does raise the true loss). Layer 0 additionally has a logging issue in this column, so we use its final-loss comparison alone. The regeneration claim rests only on the final-loss comparison after 200 further steps.

At $n = 30$, all four layers retrain to control-level final loss after being reset ($p > 0.17$ for all). The completeness values cluster near 1.0 (0.981–1.046).

After a layer is reset to random weights and retrained for 200 steps, the model reaches the control final loss (within a tiny residual at $n = 300$). This is a statement about the whole model’s endpoint, not about the reset layer. We have no evidence that the rebuilt layer takes on the *same functional*

role as the original, only that the model as a whole reaches the same loss; separating the two would need a representational comparison of the old and new layer, which we did not do. At $n = 300$, small residual deficits appear ($d = 0.12$ to 0.17), graded by layer position, all within practical tolerance.

Free lunch: In the informal sense of Section 1, the endpoint is reached again after a layer is destroyed. It costs 200 further training steps, and “functional recovery” overstates what a scalar shows: what recovers is the loss, not a verified functional role.

Classification: Final loss returns to control after a layer is reset and retrained; whether the same functional role is restored is untested.

4.11 Experiment 11: Transplantation: Context-Dependent Roles

At $n = 3$: The signal was ambiguous. A transplant advantage appeared possible.

At $n = 30$: The transplant null result resolves clearly. There is no advantage to a structured (donor) layer over a random replacement.

At $n = 300$: The null holds across all layers with no exceptions. Layer-specific p -values: L0 $p = 0.29$, L1 $p = 0.44$, L2 $p = 0.98$, L3 $p = 0.91$. Overall $p = 0.76$. The transplant null is robust: donor layers confer no advantage over random reinitialization at any layer position, even at high statistical power.

Table 11. Transplant vs. random reset ($n = 30$, 200 + 200 steps).

Layer	Transplant Final	Random Final	Gap (rand – trans)	p
L0	2.470	2.467	-0.0033	0.566
L1	2.459	2.459	+0.0008	0.804
L2	2.459	2.456	-0.0025	0.568
L3	2.454	2.457	+0.0036	0.420
Overall	,	,	+0.0003	0.880

Transplanted layers and randomly-reset layers converge to the same final loss ($p = 0.880$ overall). There is no transplant advantage, a layer from a separately-trained donor network is accepted no better and no worse than a random replacement. The network does not recognize or benefit from the structure of the donor layer; it simply rebuilds whatever is placed there.

What this shows is narrower than “the pattern is positional.” After 200 steps of retraining, a donor layer and a random layer leave the model at the same final loss, so the donor’s learned structure confers no measurable head start on this task. It does not follow that the substrate is irrelevant to the solution reached: we did not check whether donor-initialized and random-initialized runs end in the same representation, and the null may also reflect that both models learned the same task on the same data, so the donor carries little that a random layer cannot quickly rediscover. A donor trained on a different task would be a stronger test.

Classification: No transplant advantage: donor and random layers reach the same final loss on this task; whether the substrate leaves any trace in the solution is untested.

4.12 Experiment 12: Competing Objectives: Pattern Corruption

At $n = 3$: The signal was ambiguous. The distinction between adversarial and inactive layers appeared but was underpowered.

At $n = 30$: The distinction resolves sharply. Adversarial layers degrade dramatically; inactive layers are tolerated.

At $n = 300$: The adversarial degradation strengthens: competing objectives degrade by +26.3% ($p < 0.0001$, $d = +0.531$), up from +24.8% at $n = 30$. High variance persists (std = 1.31). Frozen layers remain non-significant (-0.1% , $p = 0.41$). The competing vs. freeze comparison yields $p < 0.0001$ ($d = +0.535$).

Table 12. Competing objectives ($n = 30$, 200 + 200 steps).

Condition	Final Loss	vs Control $\Delta\%$	p (vs ctrl)
Control (400 steps normal)	2.457 ± 0.341	,	,
Competing (negate L2-3 grads)	3.067 ± 1.015	+24.8%	<0.001***
Freeze L2-3	2.461 ± 0.346	+0.2%	0.462

Negating gradients for layers 2-3 causes significant degradation (+24.8%, $p < 0.001$, $d = +0.689$) with high variance (std = 1.01). But merely freezing those same layers causes negligible degradation (+0.2%, $p = 0.462$, $d = +0.136$). Competing vs. freeze: $p < 0.001$, $d = +0.693$.

This is the paper’s cleanest result. After 200 steps of normal training, freezing layers 2-3 for a further 200 costs almost nothing, while negating their gradients over the same span degrades final loss by about a quarter. The contrast is between a component that stops learning and one that is driven the wrong way, and it replicates across $n = 30$ and $n = 300$. Two cautions on wording. A frozen layer is not silent or absent: it keeps applying its trained forward map, and layers 0-1 go on training around it, so “silence” here means no further weight updates. And negation is the maximally adversarial case, gradient ascent on the loss, so a large cost is expected on its own; what the experiment adds is the size of the gap against the frozen baseline. With that understood: silence is tolerated; sabotage is catastrophic. The asymmetry is large, about +0.2% versus +24.8% at $n = 30$ (+26.3% at $n = 300$).

Classification: A layer that stops learning is tolerated (freeze); one driven by inverted gradients is not (negation). In the framework’s terms, an interface can fail to transmit or can actively distort, with very different consequences.

4.13 Cross-Experiment Synthesis

Table 13a. Perturbation tolerance (Experiments 1-6, paired t -tests, $n = 30$).

Perturbation	Final Loss $\Delta\%$	p (final)	Mean Loss		Classification
			$\Delta\%$	p (mean)	
Freeze 1-2 heads	-0.1-0.3%	0.025-0.253	-0.0%	0.002-0.053	Pattern manifestation
Freeze 4-16 heads	-0.4-0.7%	0.001-0.322	-0.1-0.3%	<0.001	Free lunch (trajectory)
Window=4,8	-0.1-0.3%	>0.30	-0.0-0.1%	0.037-<0.001	Pattern fidelity
Partial flow (25-75%)	$\pm 0.1\%$	0.379-0.915	$\pm 0.0\%$	>0.18	Pattern fidelity
Noisy $\sigma=0.01$	+0.1%	0.843	+0.0%	0.333	Pattern fidelity
Noisy $\sigma=0.1$	+2.4%	0.032	+2.4%	<0.001	Fidelity exceeded
Quantized 3-level	+3.8%	0.008	+2.5%	<0.001	Fidelity exceeded
Sign-only	+5.0%	0.002	+3.9%	<0.001	Fidelity exceeded
Cell-view (local loss)	-0.9%	0.237	+0.8%	<0.001	Final-loss invariance

Table 13b. Multi-phase perturbation results (Experiments 7-12, paired t -tests, $n = 30$).

Experiment	Condition	vs Control	p	Classification
7: Recovery	Damaged then recovered	+0.0%	0.886	Pattern re-binding
8: Chimera (AABB)	Frankenstein assembly	-0.4%	0.265	Pattern invariance
8: Chimera (ABAB)	Alternating layers	-0.0%	0.985	Pattern invariance
9: Sudden full (dose 2.0)	$\sigma = 0.1$ all steps	+2.4%	0.032*	Fidelity exceeded
9: Sudden half (dose 1.0)	$\sigma = 0.1$ from step 100	+1.0%	0.040*	Fidelity exceeded
9: Gradual (dose 0.67)	Ramp 0 to 0.1	+0.3%	0.641	Lower dose, less degradation
9: Gradual vs sudden half	Dose-matched contrast	-0.7%	0.355	No inoculation (null)
10: Regeneration (any layer)	Reset then retrain	+0.1-0.4%	>0.17	Functional role pattern

Experiment	Condition	vs Control	p	Classification
11: Transplant vs random	Donor vs random layer	$\pm 0.0\%$	0.880	Context-dependent role
12: Adversarial L2-3	Negate gradients	+24.8%	$<0.001^{***}$	Pattern corruption
12: Freeze L2-3	Zero gradients	+0.2%	0.462	Pattern fidelity

Classification under the pattern-access framework (full):

Endpoint effects the objective did not ask for, in the informal free-lunch sense of Section 1 (each paid for by extra steps, parameters, or overparameterization): - **Complete recovery (Exp 7)**: After a transient, mild freeze is lifted, final loss returns to the control level. The freeze is near-harmless on its own, so this is removal of a benign constraint more than repair. - **Regeneration (Exp 10)**: A layer reset to random weights retrain, over 200 steps, to control-level final loss. What recovers is the model’s loss; whether the same functional role returns is untested. - **Head-freezing trajectory (Exp 1)**: Frozen random-projection heads leave the endpoint unchanged and slightly improve the training trajectory, plausibly by reducing gradient interference.

Final-loss invariance, the same endpoint loss is reached through different interfaces (an equality of loss, not a demonstrated shared basin): - **Chimera convergence (Exp 8)**: Models assembled from parts of two networks retrain to control-level final loss; no assignment prevents it. - **Transplant indifference (Exp 11)**: Donor and random layers reach the same final loss, so a donor confers no measurable advantage on this task. - **Cell-view convergence (Exp 2)**: Local layerwise loss reaches the same final loss as end-to-end backpropagation, using extra projection heads and globally-supplied labels.

Pattern fidelity, the interface tolerates degradation up to a threshold: - **Gradient fidelity threshold (Exp 3)**: Small noise tolerated; large noise exceeds fidelity. - **Noise dose-response (Exp 9)**: Final-loss degradation rises with total injected gradient-noise variance; the dose-matched gradual-vs-sudden contrast shows no exposure-history (inoculation) effect. - **Layerwise autonomy (Exp 5)**: 25–75% gradient flow tolerated without meaningful degradation. - **Pattern visibility (Exp 4)**: All window sizes tolerated for final loss.

Pattern corruption, the interface actively inverts the pattern: - **Absence vs. corruption (Exp 12)**: Frozen layers tolerated ($p = 0.41$ at $n = 300$); adversarial layers degrade +26.3% ($p < 0.0001$). Silence is tolerated; sabotage is catastrophic.

DG does not track perturbation. At $n = 3$, the DG Index appeared to scale with perturbation severity. At $n = 30$, this signal resolved to null: no perturbation condition produces a statistically significant DG change. Baseline $DG = 0.680 \pm 1.131$; all conditions are non-significant ($p > 0.19$ for all). At $n = 300$, the DG null holds ($p > 0.23$ for all): the metric still does not track perturbation severity, confirming that DG captures intrinsic stochastic structure of loss trajectories rather than perturbation response.

4.14 Findings: What Faultization Revealed

Free Lunches **Finding 1: Pattern re-binds after disruption (Exp 7).** A model damaged during training (8 frozen heads for 100 steps) recovers to identical final loss at $n = 30$ ($p = 0.886$, ratio 1.0000). All 30 runs recovered within a mean of 1 step. At $n = 300$, a tiny but significant residual emerges ($p = 0.030$, $d = +0.126$) but remains within practical equivalence. Free lunch: complete recovery without a recovery mechanism.

Finding 2: A reset layer retrains to the control loss level (Exp 10). Any single layer can be reset to random weights and retrained to no detectable difference in final loss at $n = 30$ ($p > 0.17$ for all layers), with tiny residuals at $n = 300$ ($d = 0.12$ to 0.17). What returns is the model’s loss; whether the same functional role returns is untested. The 200 retraining steps are the cost.

Finding 3: Interface simplification improves trajectory (Exp 1). Freezing 4+ randomly-initialized heads produces small but statistically robust mean-trajectory improvements (freeze 8: $p < 0.001$, $d = -1.228$; freeze 12: $p < 0.001$, $d = -1.366$). Frozen random-projection heads reduce gradient interference. At $n = 30$, some final-loss improvements appear significant, but these resolve to null at $n = 300$. Free lunch: trajectory improvement without paying for it.

A Confound Caught Under Dose Control **Finding 4: No stress inoculation once noise dose is matched (Exp 9).** A naive gradual-vs-sudden comparison looks like inoculation: the gradual ramp shows no significant final-loss degradation ($+0.3\%$, $p = 0.641$ at $n = 30$) while always-on noise degrades ($+2.4\%$, $p = 0.032$), and the direct contrast is significant ($p = 0.032$, $d = -0.412$ at $n = 30$; $p < 0.0001$, $d = -0.374$ at $n = 300$). But the gradual ramp injects only about one-third the cumulative gradient-noise variance of always-on noise. When dose is matched (gradual vs. sudden half), the difference is not significant (-0.7% , $p = 0.35$ at $n = 30$; $+0.3\%$, $p = 0.20$ at $n = 300$), and degradation across the three perturbed conditions tracks total dose, not exposure history. We report this as a confound, not a free lunch: there is no evidence that the order of noise exposure, holding dose fixed, changes which patterns are accessible.

Pattern Invariance **Finding 5: Every chimera retrains to the control loss level (Exp 8).** Models assembled from parts of two independently-trained networks reach the same final loss as undamaged continuation ($p > 0.07$ for all chimera types). No layer assignment prevents it; whether the assemblies settle into the same loss-basin is not tested.

Finding 6: No transplant advantage (Exp 11). Transplanted layers and randomly-reset layers reach the same final loss ($p = 0.880$ overall), so a donor confers no measurable head start on this task. Whether the two end in the same representation is untested.

Pattern Fidelity **Finding 7: Discarding gradient magnitude costs most (Exp 3, 6).** Replacing the gradient with its sign, which keeps direction and drops magnitude, degrades final loss more (sign-only: $+5.0\%$ at $n = 30$, $+4.9\%$ at $n = 300$) than adding noise or scaling the gradient down. Magnitude information therefore matters here: keeping only the sign is not enough. Scaling the

gradient (partial flow) is mostly absorbed, but under Adam that is expected regardless of the layers. None of this is a controlled comparison with signSGD, which tunes its own step size.

Finding 8: Channel asymmetry in pattern access (Exp 6). In the 2×2 composite design, sign-only gradients (+5.0–5.2% final loss at $n = 300$) degrade far more than noisy $\sigma = 0.1$ gradients (+2.3%), regardless of forward perturbation type. The sign-only vs. noisy gradient contrast: -3.8% ($p = 0.022$ at $n = 30$; -2.8% , $p < 0.0001$ at $n = 300$, $d = -0.297$). The gradient channel is more sensitive to information destruction than the forward channel.

Finding 9: Silence tolerated, sabotage catastrophic (Exp 12). Frozen layers cost nothing (+0.2%, $p = 0.462$ at $n = 30$; $p = 0.41$ at $n = 300$); adversarial layers cost +24.8% at $n = 30$ ($p < 0.001$, $d = +0.689$) and +26.3% at $n = 300$ ($p < 0.0001$, $d = +0.531$). Pattern unavailability (silence) is absorbed; pattern corruption (inversion) is catastrophic.

Signals That Resolved Across Scales At $n = 3$, the following signals were ambiguous; at $n = 30$, they resolved:

- **Head freezing improves final loss:** At $n = 3$, this appeared as a possible improvement signal. At $n = 30$, some final-loss improvements appear significant (freeze 4: $p = 0.001$), but at $n = 300$ all resolve to null (all $p > 0.15$). A different, finer signal persists at all scales: mean-trajectory improvement. The picture sharpened, the robust signal is in the trajectory metric, not the final-loss metric.
- **DG scales with perturbation:** At $n = 3$, the DG Index appeared to scale with perturbation severity. At $n = 30$, this resolved to null across all conditions ($p > 0.19$). At $n = 300$, the null is confirmed: DG still does not track perturbation, establishing that this metric captures intrinsic trajectory structure rather than perturbation response.
- **Gradient degradation is neutral:** At $n = 3$, all four methods appeared neutral. At $n = 30$, sign-only, quantized, and noisy $\sigma = 0.1$ all resolved to significant final-loss degradation; only $\sigma = 0.01$ remained tolerated. The resolution was too coarse to see the effects at $n = 3$.
- **Partial communication outperforms full:** At $n = 3$, a U-shaped curve appeared. At $n = 30$, this resolved to flat (except at zero). The pilot U-shape was sampling noise.
- **Noise helps:** At $n = 3$, small noise appeared beneficial. At $n = 30$, this resolved to null ($p = 0.333$ for mean, $p = 0.843$ for final). The apparent benefit was within noise.

5. Discussion

5.1 What Faultization Reveals About Pattern Access

During standard training, the transformer’s components cooperate invisibly. Faultization, systematic perturbation of the interface, makes the system’s relationship to the latent space legible by forcing it to operate under constraint. Each experiment degrades the interface differently, and the four-category classification organizes what we observe.

Endpoint effects the objective did not ask for. Three findings describe endpoint or trajectory effects the loss did not specify. Recovery (Exp 7): after a transient, near-harmless freeze, final loss

returns to the control level ($p = 0.886$ at $n = 30$). Regeneration (Exp 10): a layer reset to random weights retrains to control-level final loss across all four positions ($p > 0.17$ at $n = 30$), though whether the same functional role returns is untested. Interface simplification (Exp 1): frozen heads leave the endpoint unchanged and improve the trajectory ($p < 0.001$, $d = -1.228$ to -1.366). A fourth candidate, stress inoculation (Exp 9), does not survive dose control and is reported as a confound, not a free lunch.

Final-loss invariance. Three comparisons, chimera assembly (Exp 8), transplant indifference (Exp 11), and cell-view (Exp 2), reach the same final loss through different interfaces. Cell-view (local loss) is not significantly different from baseline ($p = 0.237$ at $n = 30$, $p = 0.53$ at $n = 300$). On this scalar measure the endpoint does not depend on the communication structure of the interface; whether the underlying solution does is a question a single loss value cannot answer.

Pattern fidelity. The system maintains clean pattern access up to a degradation threshold. Gradient noise at $\sigma = 0.01$ is tolerated; at $\sigma = 0.1$ it exceeds fidelity. Partial communication reduction (25–75%) is tolerated. Vision restriction is tolerated. The fidelity threshold is sharp: below it, the pattern manifests cleanly; above it, access degrades monotonically.

Pattern corruption. The absence-vs-corruption distinction (Exp 12) defines the qualitative boundary. Frozen layers (pattern unavailability) are absorbed ($p = 0.462$ at $n = 30$, $p = 0.41$ at $n = 300$); adversarial layers (pattern corruption) degrade substantially (+26.3%, $p < 0.0001$ at $n = 300$). The interface can fail to transmit (silence) or can actively distort (corruption), and the consequences are qualitatively different.

5.2 What the Objective Did Not Specify

For each experiment we can separate what the objective specified (minimize loss) from what the run also exhibited. The difference is what we have been calling, informally, a free lunch. The label is loose in two ways worth stating plainly. First, it is not the No-Free-Lunch theorem of Wolpert and Macready (1997), which concerns performance averaged over problem classes; we are describing individual behaviours on one task. Second, nothing here is free of resources: each effect below is bought with architecture, overparameterization, auxiliary supervision, or additional training steps. “Free” means only that the effect is absent from the loss the optimizer was handed.

Experiment	Specified	Also observed	Paid for by
1: Head Freezing	Minimize loss	Trajectory improves with fewer DOF	Overparameterization; frozen heads stay forward-active
2: Cell-View	Local loss per layer	Same final loss	Per-layer projection heads; globally-supplied labels

Experiment	Specified	Also observed	Paid for by
7: Recovery	Minimize loss	Same final loss after a mild freeze	200 further training steps
10: Regeneration	Minimize loss	Same final loss after a layer reset	200 further steps; overparameterization

Stress inoculation (Exp 9) was previously listed here and has been removed: once total noise dose is controlled, there is no order effect to explain, so it is a confound, not a capability. The remaining experiments characterize the interface rather than adding to this list: they map the tolerance boundary (Exp 3, 4, 5), the channel asymmetry (Exp 6), the final-loss invariances (Exp 8, 11), and the freeze-versus-negation asymmetry (Exp 12). None of the four rows above, read strictly, is more than a statement about final loss reached at some resource cost; the richer readings (recovery, regeneration, global convergence) are interpretation, and are offered as such.

5.3 The Pattern-Interface Framework

On the reading we adopt, the transformer is an interface and the loss-landscape structure (its minima, barriers, and tolerance thresholds) is what training converges onto. This is a vocabulary for organizing the results, not a claim to have measured a separate realm: everything we actually recorded is final loss.

Faultization degrades the interface. The four categories capture what happens:

1. **Pattern manifestation:** the interface is degraded but the training endpoint is still reached, despite fewer parameters (Exp 1), different communication architectures (Exp 2, 5), restricted visibility (Exp 4), or chimeric construction (Exp 8).
2. **Pattern fidelity:** there exists a threshold below which the interface maintains clean access and above which it does not. The gradient noise threshold ($\sigma = 0.01$ vs. $\sigma = 0.1$ in Exp 3), the sign-structure sensitivity (Exp 6), and the communication topology results (Exp 5) map this boundary.
3. **Pattern corruption:** the interface does not merely fail to transmit; it actively inverts the signal. Gradient negation (Exp 12) is qualitatively different from gradient silencing. The pattern is not just unavailable, a corrupted version is imposed.
4. **Free lunches:** the system receives capabilities that were not specified by the optimization objective. Recovery (Exp 7), regeneration (Exp 10), and trajectory improvement (Exp 1) are free lunches. The optimizer paid for “minimize loss”; it received path-independent recovery, functional regeneration, and trajectory improvement at no additional cost. A fourth candidate, stress inoculation (Exp 9), was ruled out once total noise dose was controlled.

This framework does not claim that patterns literally reside in a separate realm. It claims that the distinction between what the optimizer specified and what the system exhibits is real, measurable,

and informative, and that the Platonic Space framework (Levin, 2026) provides a useful vocabulary for organizing these observations.

5.4 Connection to Levin’s Platonic Space

Levin (2026) proposes that non-physical patterns are real, discovered, and causal; that physical systems are interfaces through which these patterns manifest; and that evolution, learning, and engineering exploit patterns as affordances they do not need to pay for. Our findings map onto this framework as follows.

“Non-physical patterns are real” maps to loss-landscape structure. The final-loss level that the perturbed runs reach is a property of the task and its loss landscape, not of any single weight configuration. Different interfaces (chimeras, cell-view, head-frozen, transplanted) reach the same final loss. This is consistent with the framework’s reading, but reaching the same loss is not the same as accessing the same pattern, and we have not shown the stronger claim.

“Physical systems are interfaces” maps to transformer weights. The weights, gradients, and communication channels are the physical substrate through which the optimization pattern manifests. Faultization degrades this substrate. The pattern manifests through the degraded interface up to a fidelity threshold, beyond which access is lost (Exp 3) or corrupted (Exp 12). The interface is not the pattern, it is the channel through which the pattern is accessed.

“Free lunches” map to recovery, regeneration, and trajectory improvement. The optimizer specifies loss minimization. It does not specify path-independent recovery (Exp 7), functional regeneration (Exp 10), or trajectory improvement under interface simplification (Exp 1). These capabilities appear without being paid for by the optimization process. In Levin’s framework, they are patterns from the latent space that the physical process of gradient descent exploits as affordances. We had counted stress tolerance (Exp 9) among these, but dose control removed it: what looked like a free lunch was the gradual condition simply receiving less noise.

The strength of this mapping should not be overstated. The Platonic Space interpretation is a framework for organizing findings, a way of asking “what did the system receive that it did not pay for?”, rather than a falsifiable metaphysical claim about the ontological status of mathematical objects. What is falsifiable is the predictions it generates: that systematic perturbation of any sufficiently capable interface should reveal free lunches (capabilities not specified by the construction or training process), and that these free lunches should have a structured relationship to the interface’s architecture. Our twelve experiments are consistent with these predictions.

5.5 Connection to Distributed Chess

Kofman, Campitelli & Levin (2025) implemented a distributed form of chess where each piece operates as an autonomous agent. Experiments 4-6 tested three predictions; Experiments 7-12 extend the morphogenetic paradigm beyond the chess paper’s framework.

Information bottleneck as beneficial constraint (partially supported at $n = 300$). The chess pa-

per’s central result, intermediate vision radius $R4$ outperforms omniscient $R7$, does not translate to attention windowing at $n = 30$ (all final-loss $p > 0.30$). At $n = 300$, however, fine structure emerges: window 1 significantly harms ($p = 0.021$, $d = +0.134$), window 8 significantly improves ($p = 0.022$, $d = -0.133$), and intermediate windows are neutral. This monotonic structure is consistent with a weak pattern visibility effect that was below detection threshold at lower power.

Partial communication tolerance (confirmed as pattern fidelity, not improvement). Reducing gradient flow to 25% produces no significant degradation, but partial flow does not *improve* over full backpropagation. The pattern fidelity is real; the U-shaped curve from $n = 3$ pilot data was noise.

Courage/caution strategy (channel sensitivity). In the 2×2 factorial design, gradient type is the primary driver: sign-only gradients (a, c) degrade final loss by +5.0–5.2% at $n = 300$, while noisy $\sigma = 0.1$ gradients (b, d) degrade by +2.3%. The sign-only vs. noisy gradient contrast: -3.8% final ($p = 0.022$ at $n = 30$; -2.8% , $p < 0.0001$ at $n = 300$); -1.5% mean ($p < 0.001$, $d = -1.921$). The gradient channel is the critical interface signal for pattern access.

Stress inoculation (tested, ruled out, Exp 9). We tested whether gradual noise exposure builds tolerance that sudden exposure does not. The unmatched gradual-vs-sudden-full contrast is significant ($p < 0.0001$ at $n = 300$, $d = -0.374$), but it confounds exposure history with total noise dose: the gradual ramp delivers about one-third the cumulative gradient-noise variance of always-on noise. The dose-matched contrast (gradual vs. sudden half) is not significant at either scale. We therefore find no evidence of an inoculation effect in this substrate. Whether a properly dose-matched protocol would reveal one at longer horizons, or in chess or biological systems, remains open.

Chimera convergence (new, Exp 8). Unlike biological chimeras, which can develop abnormally at graft boundaries, transformer chimeras converge seamlessly. This reflects the smoothness of the loss landscape versus the discrete developmental signaling in biological systems, the interface is simpler, and the pattern invariance is correspondingly wider.

5.6 Scaling as Methodology

The three-scale protocol is not merely a replication strategy, it is a methodological commitment. At $n = 3$, the resolution is coarse. Real effects can be invisible; noise can masquerade as signal. At $n = 30$, the picture sharpens: moderate effects become detectable, and many $n = 3$ ambiguities resolve. At $n = 300$, fine structure emerges that $n = 30$ cannot see.

The practical implication: do not read findings at $n = 3$, or the borderline ones at $n = 30$, as conclusions. The head-freezing final-loss improvements that looked significant at $n = 30$ (freeze 4: $p = 0.001$) dissolved at $n = 300$ (all $p > 0.15$), and the honest description is that they were small-sample false positives. What survived at all scales was a different, trajectory effect. The picture changed because resolution increased and the underpowered effects washed out, not because a real low-resolution finding was retracted.

The $n = 300$ results confirm this framing. Some non-significant findings at $n = 30$ revealed fine structure at $n = 300$ (vision radius, regeneration layer specificity). Some robust findings at $n = 30$ strengthened (sign-only degradation from $p = 0.002$ to $p < 0.0001$; head-freezing trajectory improvement strengthened in effect size). Some $n = 30$ final-loss significances resolved to null at $n = 300$ (head-freezing final loss), clarifying that the trajectory metric captures the real phenomenon. The scale did not, however, rescue the gradual-vs-sudden result: the larger sample made the dose-confounded contrast more significant, not more interpretable, and the dose-matched contrast stayed null at both scales. More data sharpens a confounded comparison without resolving the confound; only the dose-matched design does that. The three-scale protocol still achieved its design objective: coarse signal at $n = 3$, resolved signal at $n = 30$, fine structure at $n = 300$.

5.7 Limitations

Scale. The model has 4 layers, 16 dimensions, and $\sim 13,400$ parameters. Whether these findings extend to production-scale transformers is unknown. The free lunches (recovery, regeneration, trajectory improvement) may be specific to small models with simple loss landscapes, or they may be architectural universals.

Task complexity, and missing baselines. Character-level name generation is a toy task with a 28-character vocabulary and a 13.4k-parameter model, and the window-1 result suggests it needs little long-range context. Much of the apparent invariance may therefore reflect task triviality rather than a deep property of transformers: if a bigram or an attention-free model nearly matched this network, the right conclusion would be that the task does not strongly constrain the interface. We did not run those baselines (n-gram, MLP, attention-free), and without them the invariance results should be read as facts about this task, not about transformers in general.

Scalar outcome, unmeasured solutions. Every result is a single scalar, final training loss. We did not measure whether interfaces that reach the same loss occupy the same loss-basin, compute the same function, or use the same representation. Claims of that kind (basin universality, shared functional role, substrate irrelevance) would require interpolation and mode-connectivity analysis (Garipov et al., 2018), representational-similarity measures (Kornblith et al., 2019), or model stitching (Bansal et al., 2021), none of which we ran. Where earlier drafts used such language, this version reports equalities of endpoint loss instead.

Equivalence not formally tested. The null-like results (recovery, chimera, transplant, freeze) are reported as “no detectable difference in final loss,” which is not the same as demonstrated equivalence. A proper equivalence claim would predefine a smallest effect of interest and run two one-sided tests (Lakens et al., 2018); we did not.

Optimizer and ablation controls. Several gradient results are entangled with the fixed Adam recipe. Scaling gradients (Exp 5) is close to a no-op under Adam’s rescaling invariance, and sign-only updates (Exp 3, 6) are not the signSGD algorithm, which tunes its own step size. A clean read would repeat these with plain SGD and momentum SGD. Likewise, “freezing” a layer stops

its updates but leaves it forward-active; a true silence condition (zeroing or bypassing the layer) was not tested, so the freeze-versus-negation contrast is no-further-plasticity versus adversarial plasticity, not absence versus corruption.

Training duration and dose control. 200 steps per phase captures early learning dynamics. Experiment 9’s conditions confound exposure history with total noise dose, so the experiment cannot test for inoculation as designed; only the dose-matched gradual-vs-sudden-half contrast can, and it is null. A clean test would equalize cumulative injected variance across schedules and report degradation as a function of dose. Whether any order effect exists at matched dose, or at longer horizons, remains untested here.

DG metric. The DG Index does not respond to perturbation at $n = 30$ or $n = 300$, resolving an ambiguous $n = 3$ signal.

Effect sizes. Many statistically significant effects are practically negligible ($<0.5\%$). Statistical significance at $n = 30$ does not imply practical importance.

Endpoint metric. “Final loss” is the loss at the last training step, and across seeds its standard deviation (about 0.407) is an order of magnitude larger than that of the mean-trajectory loss (about 0.028). This is consistent with final loss being a single high-variance step rather than a smoothed or held-out measure, and it is one reason several final-loss effects are small and unstable while the corresponding trajectory effects are cleaner. A fixed held-out evaluation set, or an average over the last several checkpoints, would be a more stable primary endpoint; we did not use one.

Transplant design. The null result for transplant advantage (Exp 11) may reflect that both models learned the same task on the same data. Cross-task transplantation (donor trained on a different task) might show transplant effects.

Competing objectives design. Gradient negation (Exp 12) is a maximally adversarial perturbation. Subtler forms of inter-layer conflict might reveal more nuanced compensation mechanisms.

Interpretive framework. The Platonic Space interpretation (Levin, 2026) is a framework for organizing findings, not a falsifiable metaphysical claim. The free lunches we identify are real and measurable; whether they are best understood as “patterns from a latent space” or as consequences of loss landscape geometry that we have not yet fully characterized is an open question. The framework is useful to the extent that it generates productive research questions, particularly the question of what a system receives that it did not pay for.

$n = 3$ to $n = 30$ to $n = 300$ signal evolution. Several $n = 3$ signals changed character at $n = 30$, underscoring the danger of low-power pilot data. The $n = 30$ to $n = 300$ transition showed a different pattern: no robust finding reversed, but new fine-structure signals emerged (vision radius monotonic structure, regeneration layer specificity), and several effects tightened substantially. The most instructive cross-scale change was Exp 1 head-freezing final loss, which showed several significant improvements at $n = 30$ (freeze 4: $p = 0.001$) but all dissolved at $n = 300$ (all $p > 0.15$), confirming that the trajectory metric, not the final-loss metric, captures the real phenomenon. A cautionary counterpoint is Exp 9: scaling made the dose-confounded

contrast more significant while the effect size fell (d : -0.412 to -0.374), a reminder that more samples cannot fix a design confound.

5.8 Future Work

- **Scale:** Replicate at 100M+ parameter scale to test whether the free lunches (recovery, regeneration, trajectory improvement) persist or are specific to minimal architectures.
- **Dose-matched stress inoculation:** Re-run Experiment 9 with cumulative injected gradient-noise variance equalized across schedules (gradual, sudden, and intermediate), reporting degradation as a function of dose, to test whether any exposure-order effect exists once the dose confound is removed.
- **Cross-task transplantation:** Test transplant advantage when donor and recipient are trained on different tasks.
- **Graded adversarial conflict:** Scale gradient negation from 0% to 100% to find the corruption threshold and whether free lunches appear at intermediate levels.
- **Architecture morphogenesis:** Allow the architecture to change during training, growing heads, pruning inactive ones, to test whether free lunches extend to structural self-modification.
- **Composite perturbation:** Simultaneously apply multiple perturbation types for more faithful courage/caution testing and to test whether free lunches interact.
- **Biological comparison:** Apply the same 12-experiment protocol to biological developmental systems using the same statistical framework, enabling direct cross-substrate comparison of where free lunches appear.
- **Map the pattern space:** Vary task and architecture while holding the faultization probes constant, to map which patterns are architecture-specific and which are universal. This is a direct contribution to Levin’s (2026) research program of exploring the latent space.
- **Beyond $n = 300$:** The three-scale protocol is now complete. Future replication at $n = 1000$ or with different random seeds would further constrain effect size estimates, but the primary findings are stable across all three scales.

6. Conclusion

We applied faultization, systematic morphogenetic perturbation, to a minimal transformer through twelve experiments in two phases. The central question was: when we perturb the interface, what patterns from the latent space does perturbation reveal, and what free lunches does the system receive? At $n = 300$, the picture is clear:

Effects the objective did not specify. Three findings, in the informal sense of Section 1, each bought with extra training steps or parameters. Recovery (Exp 7): final loss returns to the control level after a transient, mild freeze ($p = 0.886$ at $n = 30$, ratio 1.0000 ± 0.008 ; at $n = 300$ a tiny residual, $p = 0.030$, $d = +0.126$). Regeneration (Exp 10): a reset layer retraining to control-level final loss at $n = 30$ ($p > 0.17$), with tiny layer-specific residuals at $n = 300$ ($d = 0.12$ to 0.17). Interface simplification (Exp 1): frozen heads leave the endpoint unchanged and improve

the trajectory ($p < 0.0001$ at $n = 300$, $d = -1.228$ to -1.421); the $n = 30$ final-loss improvements were small-sample false positives. A fourth candidate, stress inoculation (Exp 9), did not survive dose control: the gradual schedule simply receives less noise, and the dose-matched contrast is null at both scales. We report it as a confound, not a discovery.

Final-loss invariance. Chimera assembly (Exp 8), transplant indifference (Exp 11), and cell-view (Exp 2) reach the same final loss through different interfaces. Local loss is not significantly different from end-to-end backpropagation ($p = 0.237$ at $n = 30$; $+0.1\%$, $p = 0.53$ at $n = 300$). The endpoint loss does not depend on the interface; whether the solution reached does is a question a scalar loss cannot settle, and we leave it open. This endpoint invariance, together with the freeze-versus-negation asymmetry below, is the paper’s most robust contribution.

Pattern fidelity. The architecture maintains clean pattern access up to a degradation threshold. Gradient noise, partial communication reduction, and vision restriction are tolerated below the threshold.

Freeze versus negation. A frozen layer is tolerated; a negated one is catastrophic. Corrupting the gradient channel costs more than corrupting the forward channel, and among gradient corruptions discarding magnitude (sign-only) costs most (-3.8% versus noisy, $p = 0.022$ at $n = 30$; -2.8% , $p < 0.0001$ at $n = 300$, $d = -0.297$).

Faultization shows how robustly this network reaches its training endpoint under interface degradation, where it fails, and where an apparent discovery dissolves under control. The optimizer specified “minimize loss.” What the runs also showed, each at some resource cost, were a return to the same loss after a mild freeze, retraining to the same loss after a layer reset, a small trajectory improvement, and the same loss from local rather than global gradients; what they did not show was stress inoculation, which we had claimed and then withdrew once dose was matched. At $n = 3$ the shapes were rough; at $n = 30$ they sharpened; at $n = 300$ some sharpened further and some dissolved. Vision radius showed only a weak, multiplicity-fragile effect, regeneration showed tiny layer-specific residuals, head-freezing final-loss significances resolved to null while the trajectory effect strengthened, and the gradual-vs-sudden contrast grew more significant without growing more interpretable. Higher resolution sharpens real structure and confounds alike; only a matched design tells them apart. What survives both the resolution dial and dose control is narrow and worth keeping: the same final loss is reached through many interfaces, a frozen layer is tolerated, and a negated one is catastrophic.

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